

WAJD CHTOUROU



ICT engineering student at SUP'COM
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SUMMARY

ICT engineering student at SUPCOM, passionate about embedded systems and telecommunication.

PROJECTS:

Job Preparation Project (P2M) : AI-Driven Spectrum Sensing & Wireless Standards Classification : Jan 2026 -May 2026

- Developed two AI-based systems for **automatic wireless communication standard identification** using Software Defined Radio:
 - Trained a **1D ResNet on the RadioML 2018 dataset** for Automatic Modulation Classification across 10 modulation schemes (93.19% accuracy), combined with a rule-based decision block using frequency and bandwidth parameters captured via **ADALM-PLUTO SDR**.
 - Built a custom RF spectrogram dataset** collected in real-world environments and trained a **2D CNN** to directly classify Bluetooth, Wi-Fi, and LTE signals (89.5% accuracy).

Tools used: Python, TensorFlow, GNU Radio, MATLAB, ADALM-PLUTO SDR.

Entrepreneurship challenge: Thneyetna

Apr 2026

- Co-founded Thneyetna, a student-focused **TAXI ride-sharing** application designed to connect university students sharing overlapping routes, enabling them to split taxi fares automatically. The app provides real-time ride suggestions and calculates the cost per passenger, **reducing travel costs by up to 40%** compared to existing platforms. Led a comprehensive **market study** targeting a pool of ~115,000 university students and ~17,000 taxi drivers in Greater Tunis, conducted a competitive analysis and defined a **business model**. We secured the **4th place** out of 27 teams and reached the **final round**.

Autonomous ADCS Monitoring System — OPS-SAT CubeSat | IEEE TSYP 13

Dec 2025

AESS Challenge

- Developing an autonomous AI-powered monitoring subsystem for ESA's OPS-SAT 3U CubeSat targeting the Attitude Determination and Control System (ADCS) for **prediction and anomaly detection**. The hardware design centered on an STM32F429 Cortex-M4 microcontroller, integrating magnetometer and photodiode sensor arrays over I2C: Trained and deployed two AI layers directly on the embedded target: a Neural Network for real-time anomaly detection (95.65% accuracy, 1 ms inference latency, 2.1K RAM footprint) and 9 Random Forest models for per-sensor signal correction. The system operates fully autonomously within tight space constraints (under 0.5 W power, 70 g, and rated for radiation and temperature extremes). We were preselected and secured 6th place in the final pitching of TSYP13 Challenge.

Summer Internship-KPIT Engineering

Jul 2025 - Aug 2025

- Developed a responsive hardware simulation of car's lights using raspberry Pi and Arduino , communicating with protocols; SOME/IP,CAN,LIN and developing a HMI using python, then Testing all the system using Robot Framework

EDUCATION

Higher School of Communication of Tunis - SUP'COM

Sep 2024 - Present

Second year ICT Engineering Student

Preparatory Institute for Engineering Studies of Sfax

Sep 2022 - Jun 2024

Mathematics and Physics field (MP)

- Ranked 138/1750 in National Exam for Engineering Schools

Mahmoud Megdich high school

Sep 2018 - Jun 2022

Baccalaureate in Mathematics

- Graduated with highest honors : 16.25 average in the national exam

ADDITIONAL INFORMATION

- Technical Skills:** Cisco Packet Tracer, Linux Red Hat Enterprise, oracle database, SQLite ,Matlab, Python, embedded c.
- Languages:** Arabic, English , French
- Certifications:**
 - TOIEC : **900/990 score (B2 level)**
 - Cisco CCNA1

- Activities:**

Community manager in IEEE GRSS chapter 2025-2026 mandate|Involved in AESS TSYP challenge|Organizing INDABAX Tunisia 2025.